

Event-Triggered Edge AI for Low-Power Wildlife & Intrusion Monitoring

Harrshetha | Easwari Engineering College

Primary SDG 15 - Life on Land

Secondary SDG 13

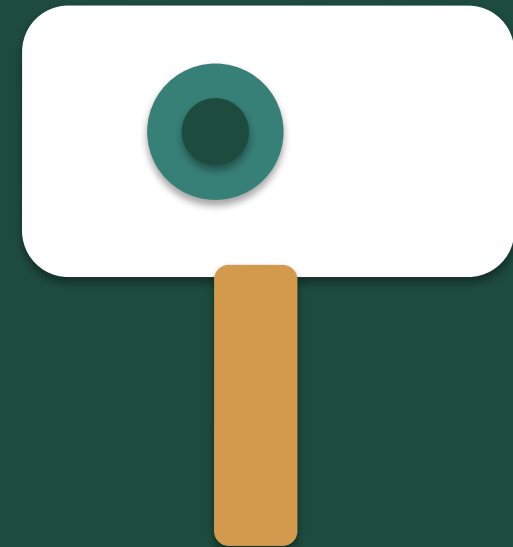
SDG 7

Event-driven sensing + on-device AI
for remote farm and forest edges

EVENT-TRIGGERED

EDGE AI

for low-power wildlife &
intrusion monitoring



Problem Statement

Start from the real constraint: remote, low-connectivity environments.

How might we use AI to detect wildlife and human intrusion in remote farm and forest edges so that conservation and crop protection can become more sustainable?

BATTERY

Continuous recording drains batteries

FOOTAGE

Useless footage overwhelms users

CLOUD

Remote areas lack connectivity for cloud AI

Design implication Do not stream everything. Trigger vision only when something changes.

Target Users

Different users need different kinds of action from the same event log.

1

FARMERS

Farmers near forest borders

Crop protection

Less nuisance; faster response when wildlife or people enter crop edges.

2

RANGE
STAFF

Forest range staff

Actionable alerts

Reduce wasted patrols by surfacing relevant events with time and class.

3

NGOs

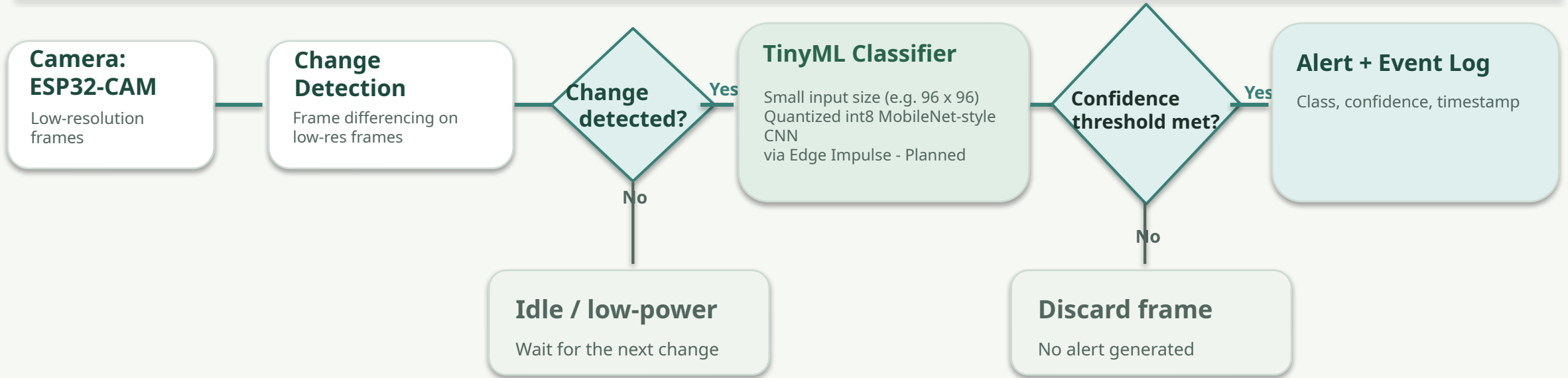
Conservation NGOs

Wildlife monitoring

Capture useful event records without continuous high-volume footage.

AI Solution Overview

A three-stage event pipeline keeps the expensive AI step conditional.



Classes

animal

human

vehicle

nothing

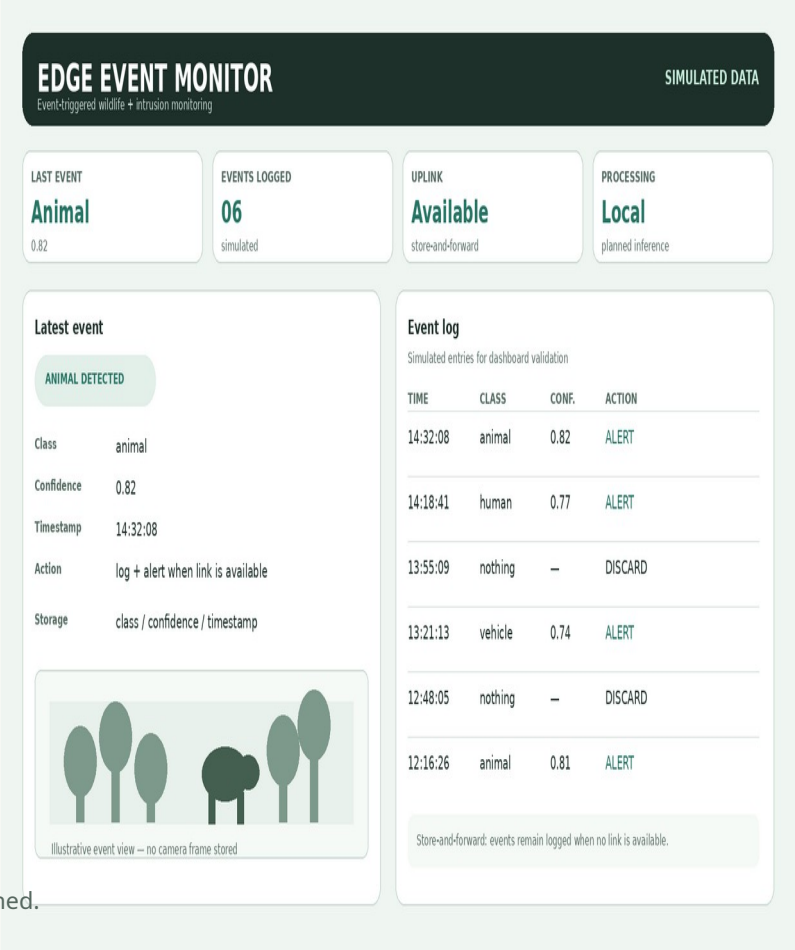
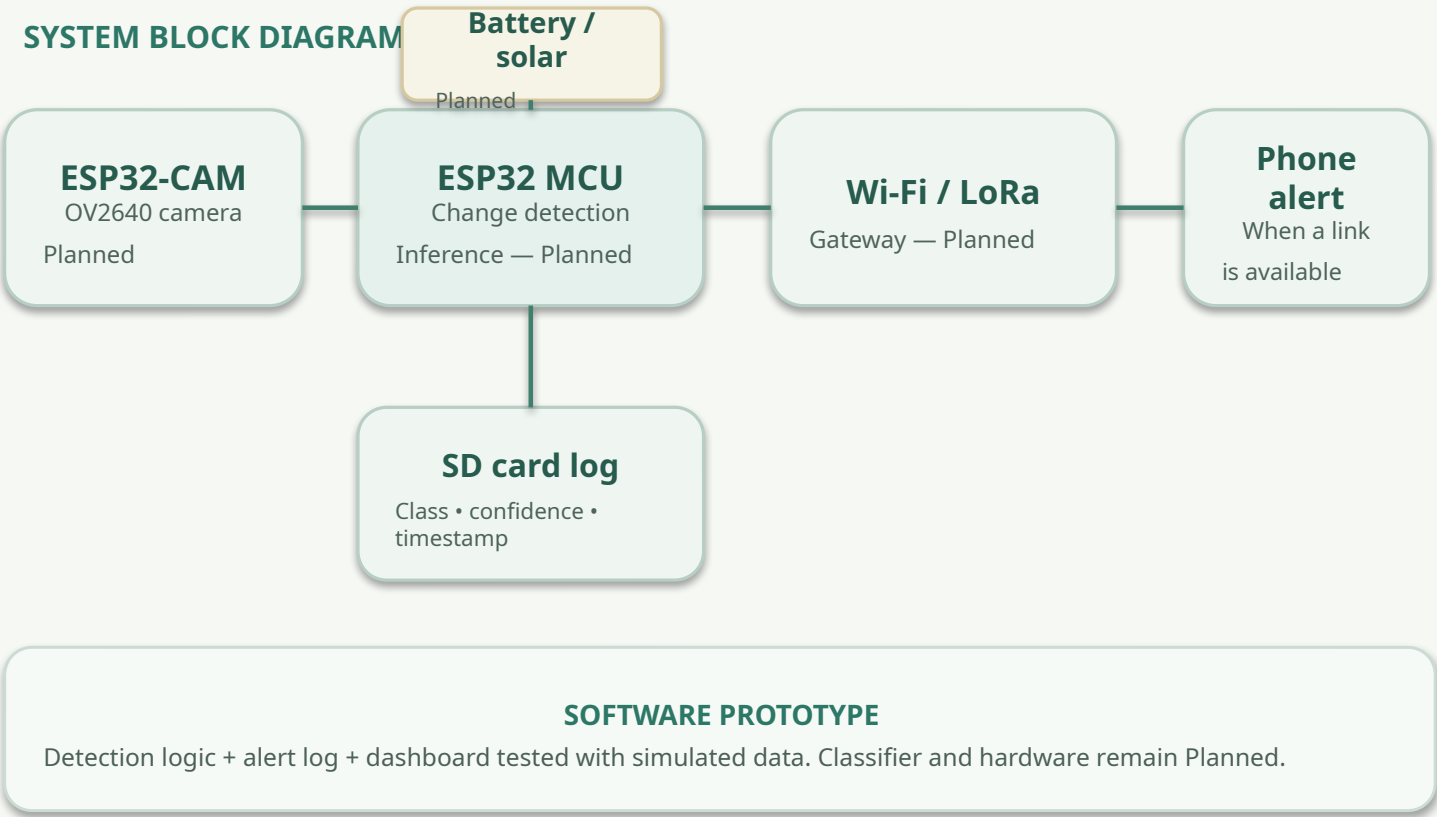
WHY AI?

Simple motion sensors can trigger on wind, shadows and vegetation. A classifier adds context - helping distinguish a person from an animal and making alerts more actionable.

05

Prototype / Demo

Software prototype: detection logic, alert log and dashboard implemented and tested on simulated data; classifier and hardware planned.



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Classifier, ESP32-CAM inference, uplink hardware and optional solar remain Planned.

Responsible AI Considerations

Responsible design is part of the system architecture, not an afterthought.

FAIRNESS

Test across lighting, weather and species so performance is not limited to ideal conditions.

TRANSPARENCY

Log every trigger with class, confidence and action; keep images only for alert events.

ETHICS

Wildlife and intrusion monitoring only; not for surveilling individuals or villages.

PRIVACY

On-device processing, discard non-event frames, no facial recognition, and no cloud upload of people images by default.

Design rule Collect the minimum data needed to act on an event.

Expected Impact

Expected outcomes (design goals) - not measured results.

DESIGN GOALS

- Processing runs only on detected change
- Fewer false alerts than motion-only triggering
- Detection and logging work offline; alerts are sent when a link is available (store-and-forward)
- Low-cost for rural deployment

WHO BENEFITS

Farmers

Less crop loss

Rangers

Fewer wasted patrols

Wildlife

Reduced human-wildlife conflict

FUTURE SCOPE

IBM Granite could summarize alert logs into plain-language daily reports for farmers or rangers - Planned.

What Changes + Next Steps

Move from continuous capture to event-driven, auditable sensing.

IF DEPLOYED

Continuous capture



Event-driven sensing

Cloud-first



Local processing

Everything becomes an alert



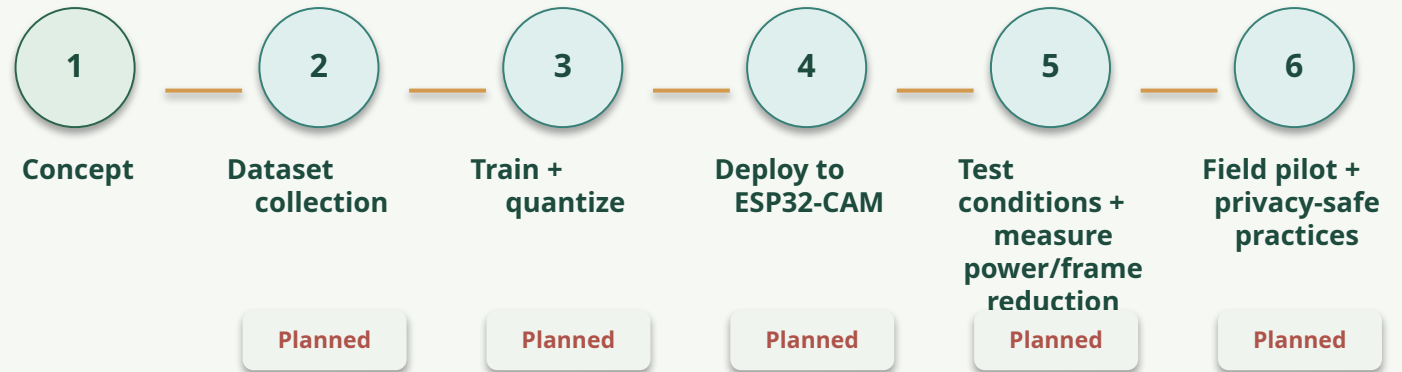
Alerts only when relevant

Opaque history



Auditable logs

ROADMAP



KEY TAKEAWAY

Sense only when needed, decide locally, and keep a traceable event record.